



From lecture session 3: We understood how the water accumulated on Earth and how it may be connected to the origin of life. Now we need to understand why it is distributed over the Earth's surface the way it is.

Lecture Session 4

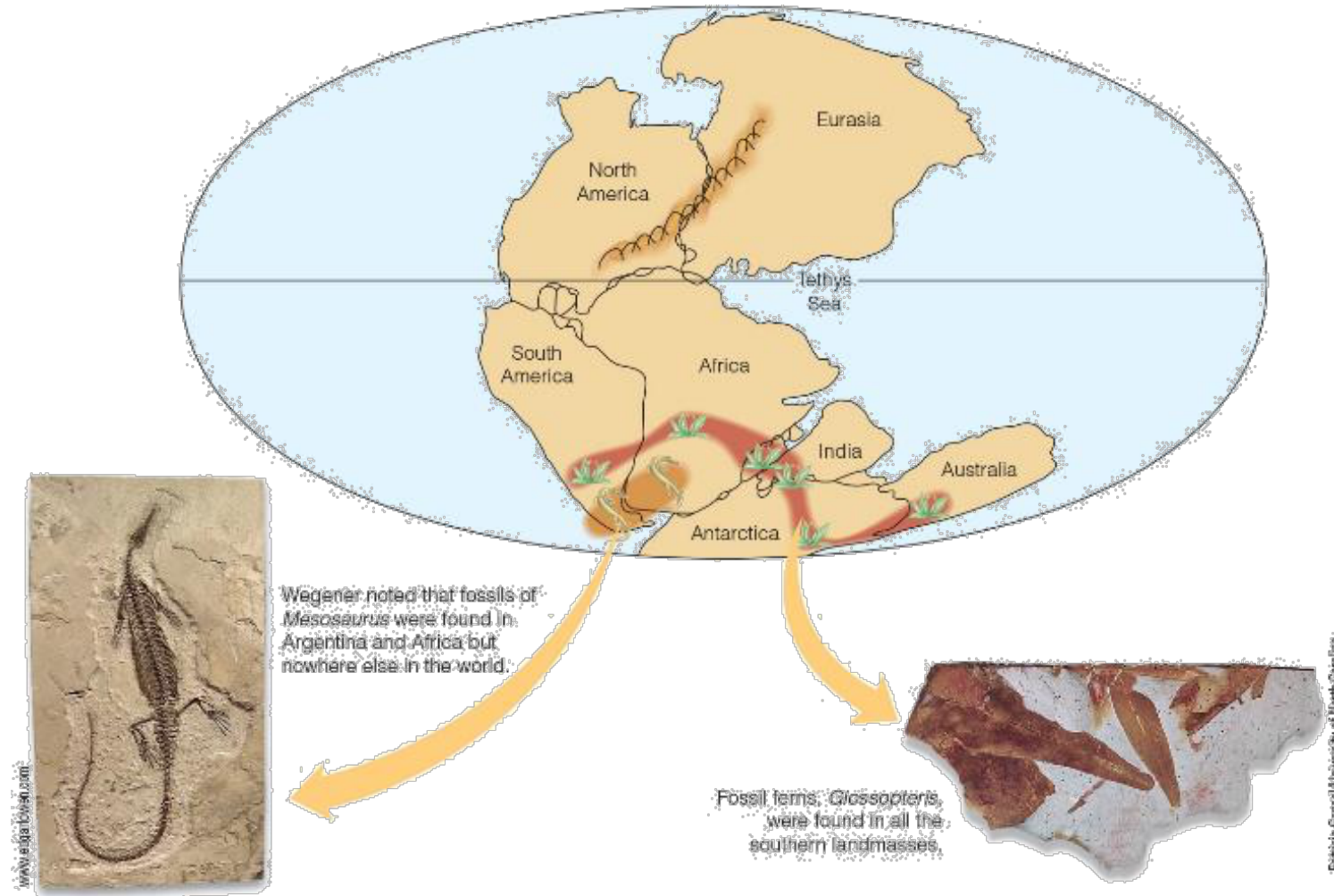
1. The continental drift theory;
2. Earth's structure;
3. Introduction to plate tectonics.



Did the oceans
always look like
this?

The continental drift theory

- Theory proposed by Alfred Wegener (1912)
 - Pangaea (continent)
 - Panthalassa (ocean)
- Evidences: shape of the continents, fossil, and rock evidence across continents
- Unexplained factors: mechanism moving the continents (Ex: centrifugal force due to Earth's rotation and/or tides); Solid support must exist below the crust to support the continents and mountains.
- Theory widely discredited at the time. Wegener was accused of “ignoring” controversial evidence to explain his own

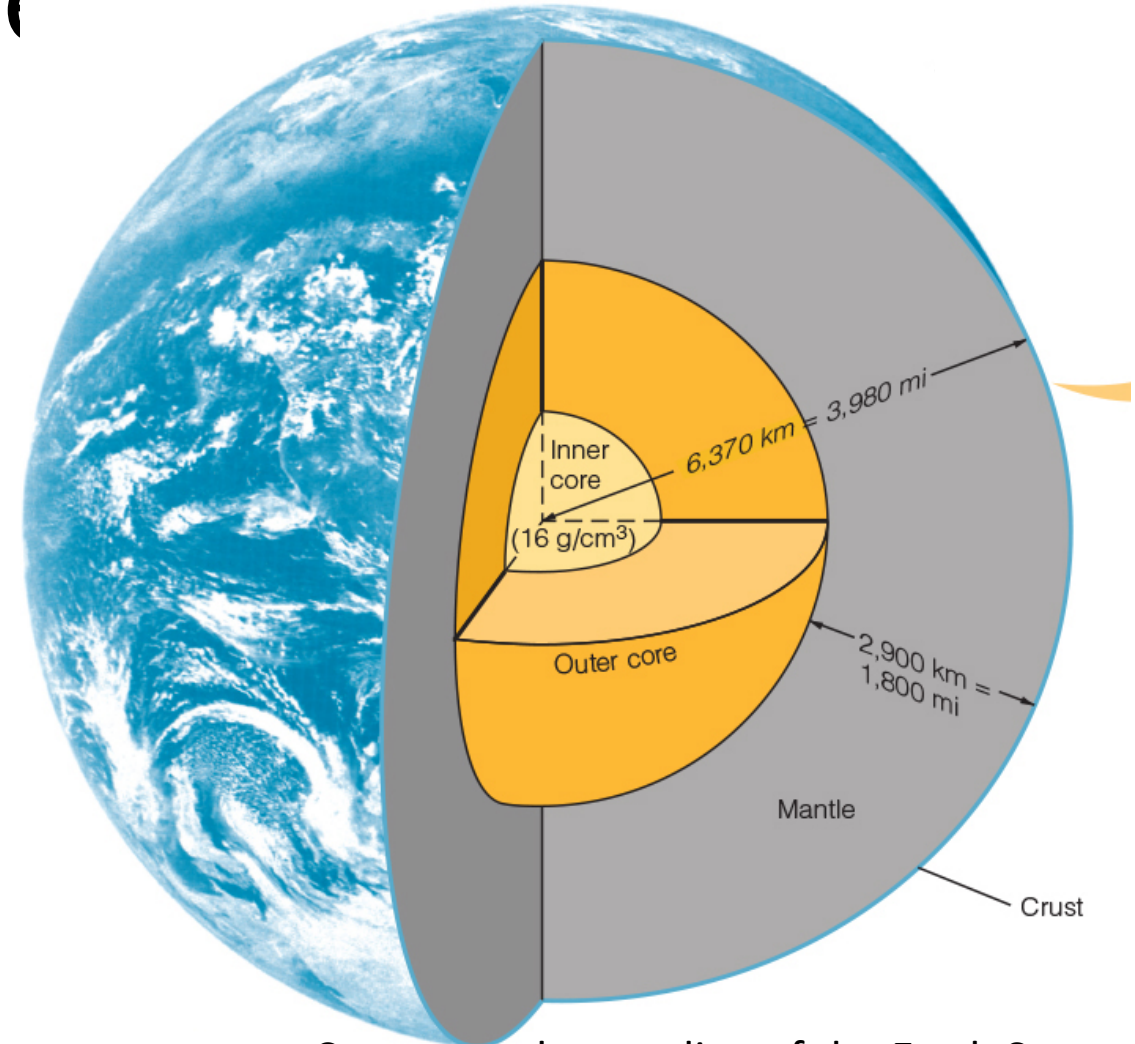


To understand the unexplained mechanisms, we need to understand the Earth's structure

Key Concept: Density = mass/volume

Remember: The Earth formed through accretion due to gravity.

- Gravity tends to “sort” denser materials below lighter ones. As a result, the Earth is a stratified (or layered) system: **Crust (solid), Mantle (mostly solid but mobile), Inner Core (liquid), Outer Core (solid).**
- The density of each layer is mainly defined by its composition, temperature, and pressure of the system.
- Our current understanding of the Earth's mainly comes from indirect measurements of seismic waves.



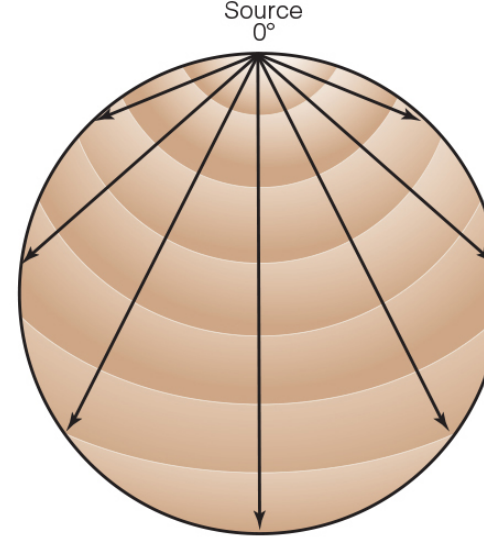
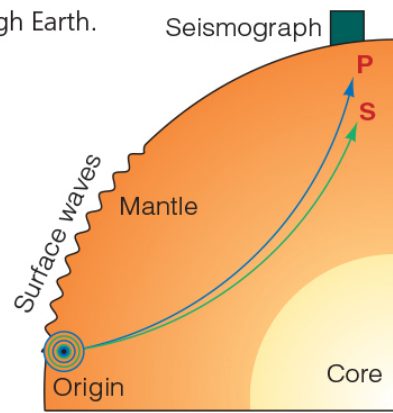
Current understanding of the Earth Structure

Earthquakes provided most of the clues

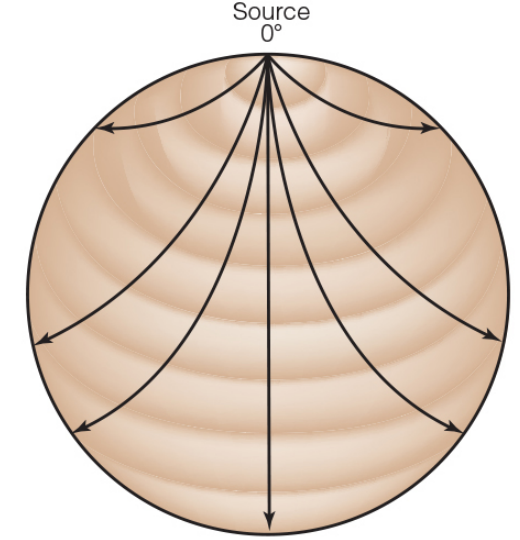
- Measurements of seismic waves using seismograph together with wave theory gave it way the dimensions and density of each layer;
- Other analyses: volcanic gases composition and variations of the Earth's gravitational pull revealed the interior's composition;

Figure 3.6 Seismic waves travel through Earth.

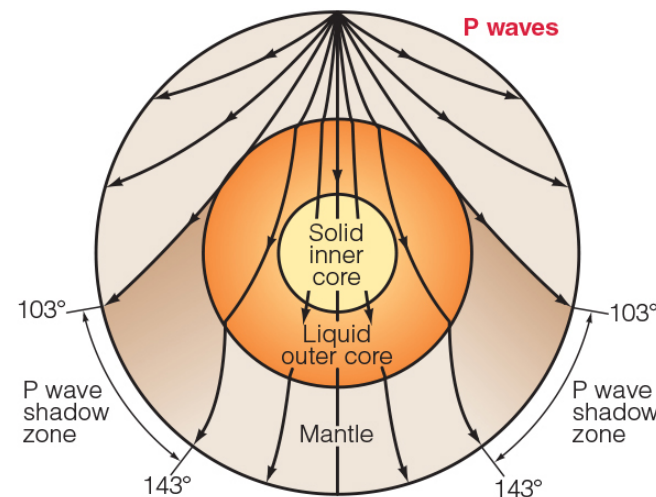
- a** A cross section of Earth shows the paths taken by surface waves and body waves (P waves and S waves). The seismographic recording shows the arrival times of waves from an earthquake located about 30° from the seismometer.



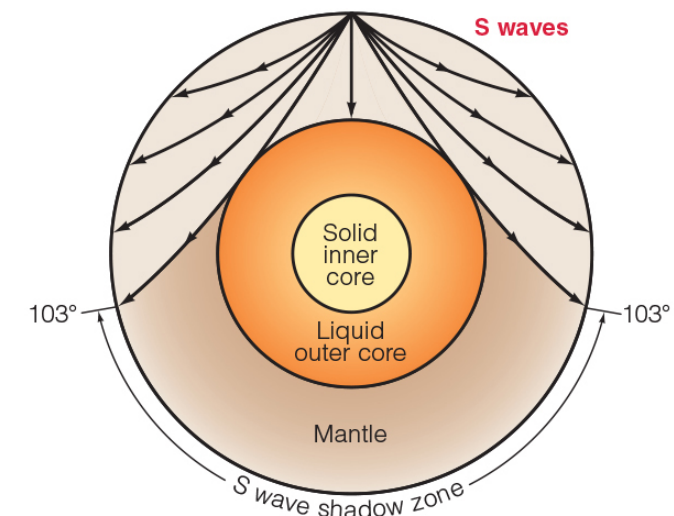
- a** Earthquake waves passing through a homogeneous planet would not be reflected or refracted (bent). The waves would follow linear paths (arrows).



- b** In a planet that becomes gradually denser and more rigid with depth, the waves would bend along evenly curved paths.



- c** P waves (compressional waves) can penetrate the liquid outer core, but are bent in transit. A P wave shadow zone forms between 103° and 143° from an earthquake's source.



- d** Our Earth has a liquid outer core through which the side-to-side S waves cannot penetrate, creating a large "shadow zone" between 103° and 180° from an earthquake's source. Very sensitive seismographs can sometimes detect weak P wave signals reflecting off the solid inner core.

Figure 3.7 How earthquakes contributed to our model of the layered Earth.

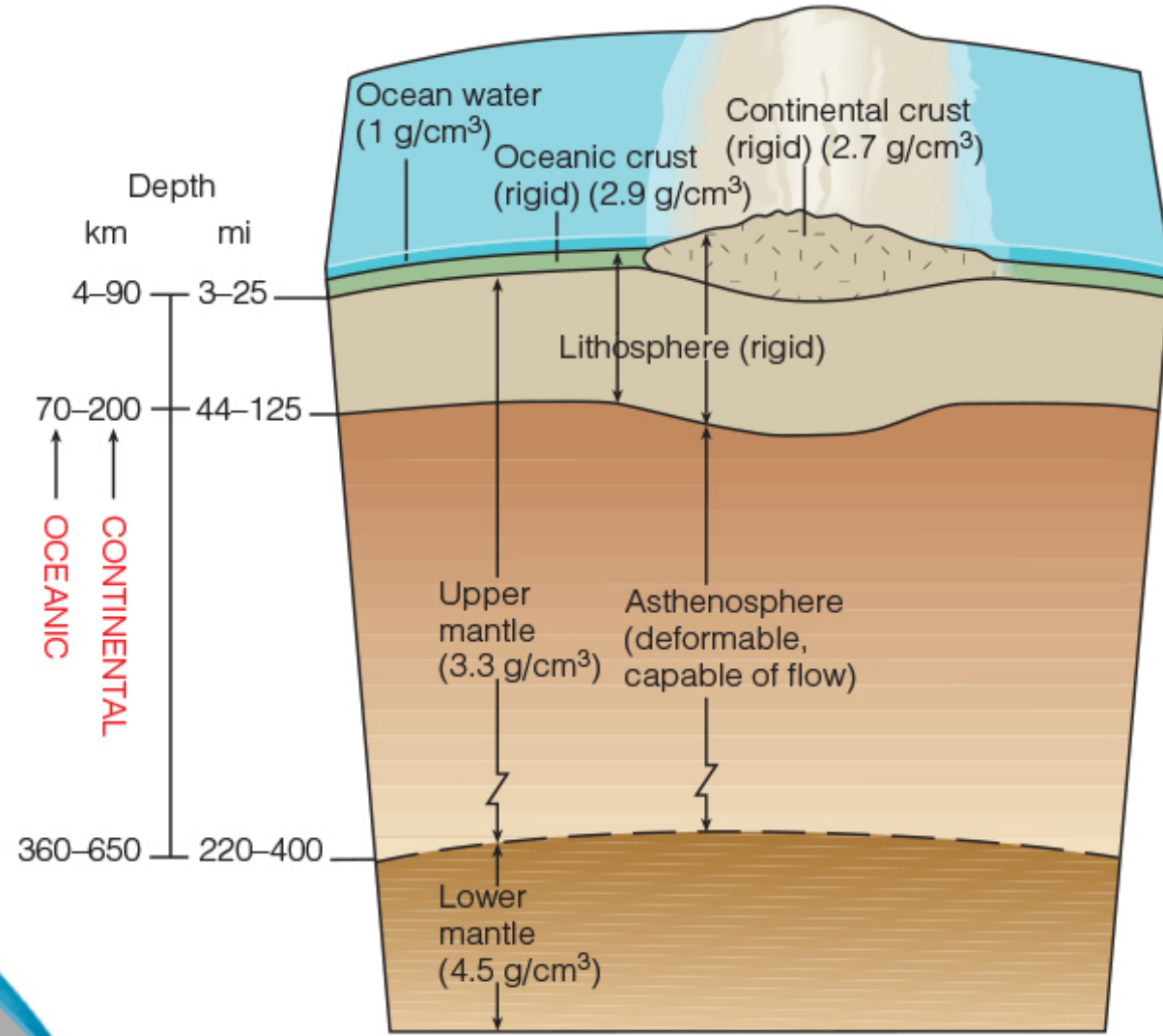
Refining our knowledge about the layers

- Earth's internal layers based on chemical composition
- Earth's internal layers based on physical properties

Refining our knowledge about the layers

- Earth's internal layers based on chemical composition
 - Crust:
 - Oceanic Crust = primarily basalt (heavy, dark-colored rock)
 - Continental Crust = primarily granite (lighter speckled rock)
 - Mantle: 68% of Earth's mass, has similar composition of the crust, but more concentration of Mg and Fe, therefore denser.
 - Core: Mostly Fe and Ni

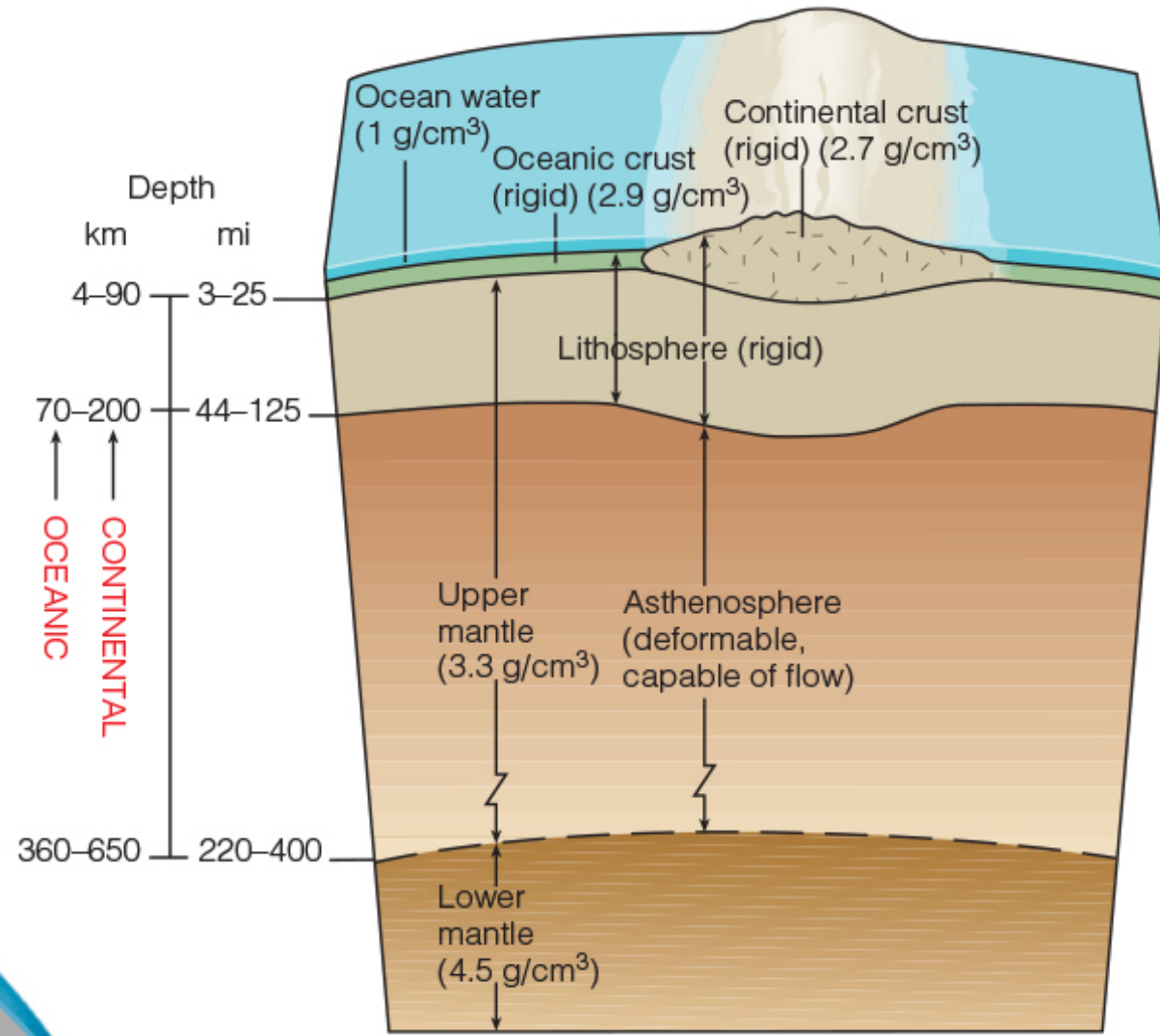
Cross-sectional view of the Earth's layers



Refining our knowledge about the layers

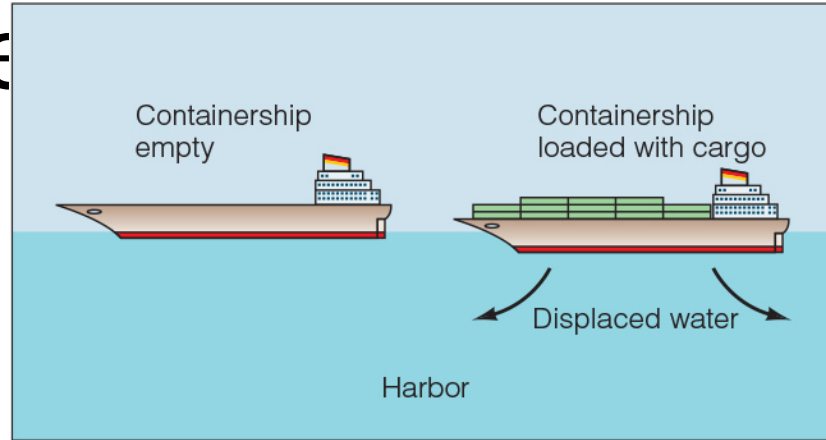
- Earth's internal layers based on physical properties
 - Lithosphere: Cool, rigid, rocky outer layer.
 - Asthenosphere: Hot, partially melted, mobile part of the upper mantle.
 - Lower mantle: Similar composition to the Asthenosphere, however it is hotter due to great pressures. Moves slower than upper mantle.
 - Core: Extremely hot and dense center of the Earth. The core accounts for nearly 1/6 of the Earth's volume, but it has 1/3 of its mass (Super Dense).

Cross-sectional view of the Earth's layers

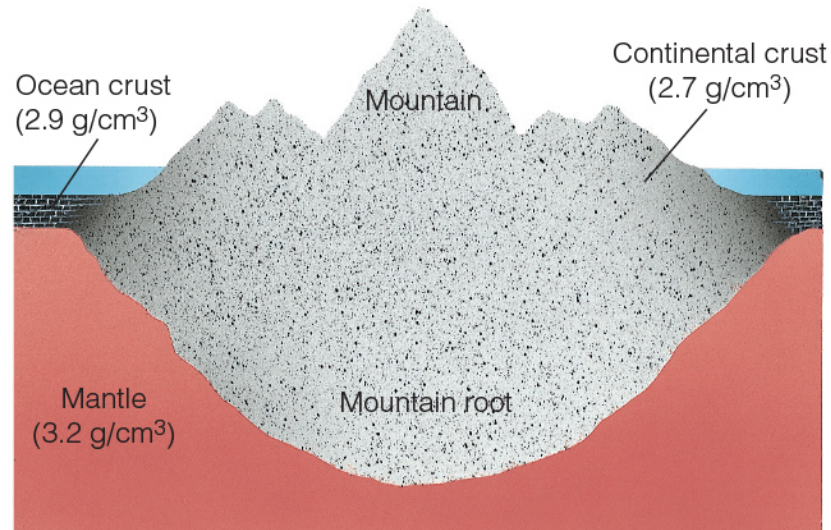


How our layers

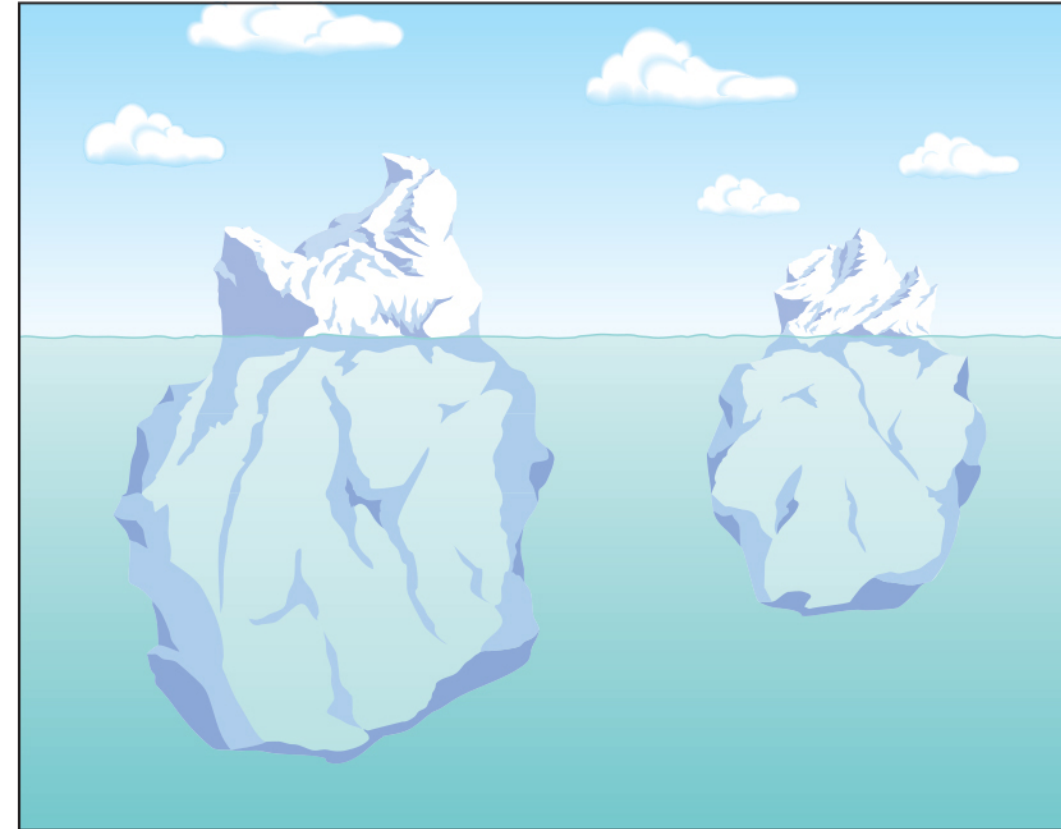
Like a ship or iceberg floating on seawater, the Earth's lightweight layers are "floating" on denser layers. Our continents and oceans (the crust) are being sustained by Earth's Mantle.



- a** A ship sinks until it displaces a volume of water equal in weight to the weight of the ship and its cargo.



- c** Continents are supported in a similar way.



- b** Icebergs sink into water so that the same proportion of their volume (about 90%) is submerged. The more massive the iceberg, the greater this volume is. The large iceberg rides higher, but also extends to a greater depth than the small one.

Figure 3.10 The principle of buoyancy and the effect of isostasy.

How the crust moves? Was Wegener right after all?

He had a great insight of the overall process, but the evidence necessary to understand it and build a more complete and comprehensive theory (**aka Plate Tectonics**) only slowly came to light between 1940s and 1970s.

Advances on the following scientific areas helped us to put the pieces together:

1. Radioactive decay: Interior heat source and incredible tool for dating rocks and sediments;
2. Earthquakes and volcanic activity maps;
3. Ocean floor mapping tools: Development of echosounders.

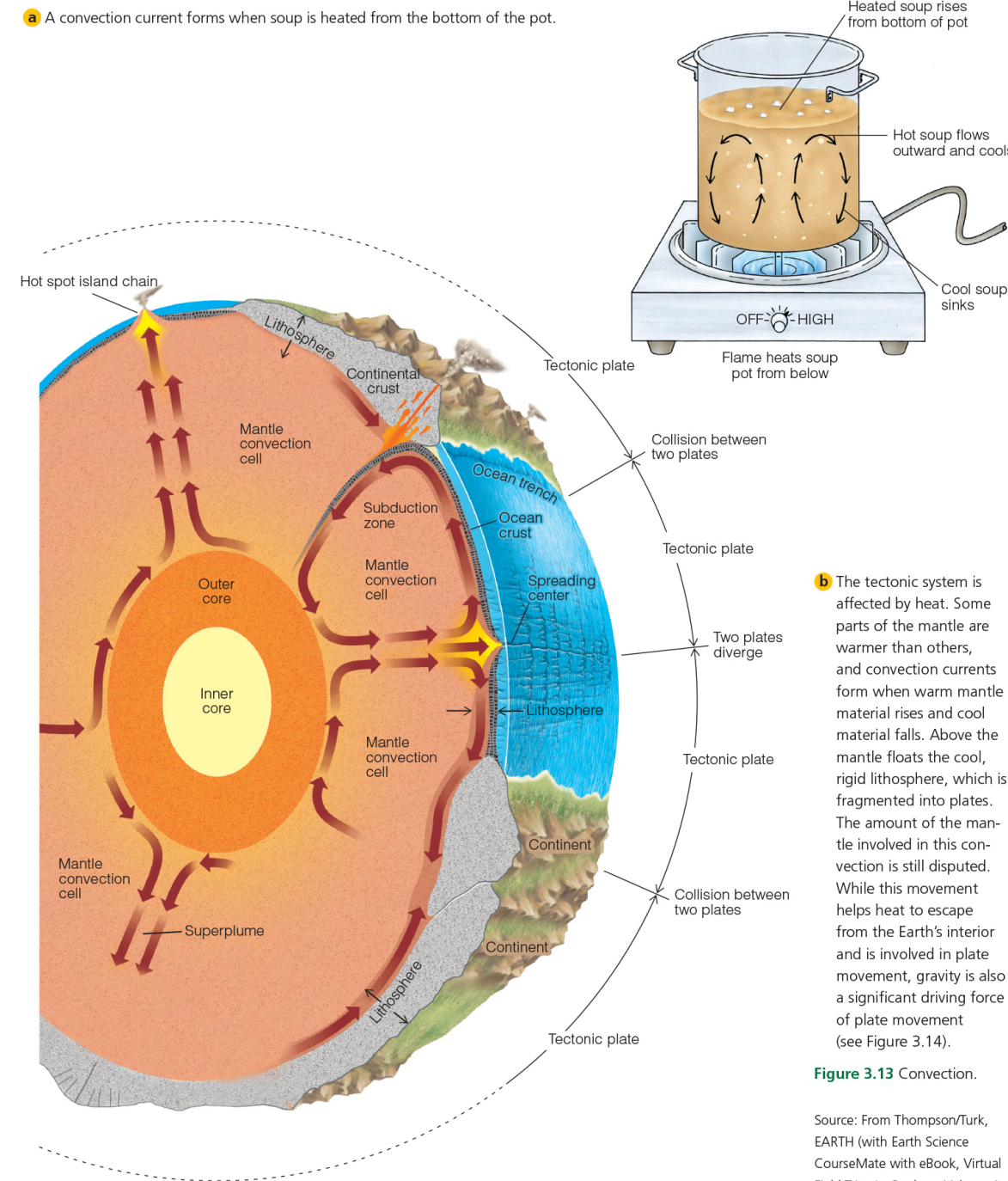


Figure 3.13 Convection.

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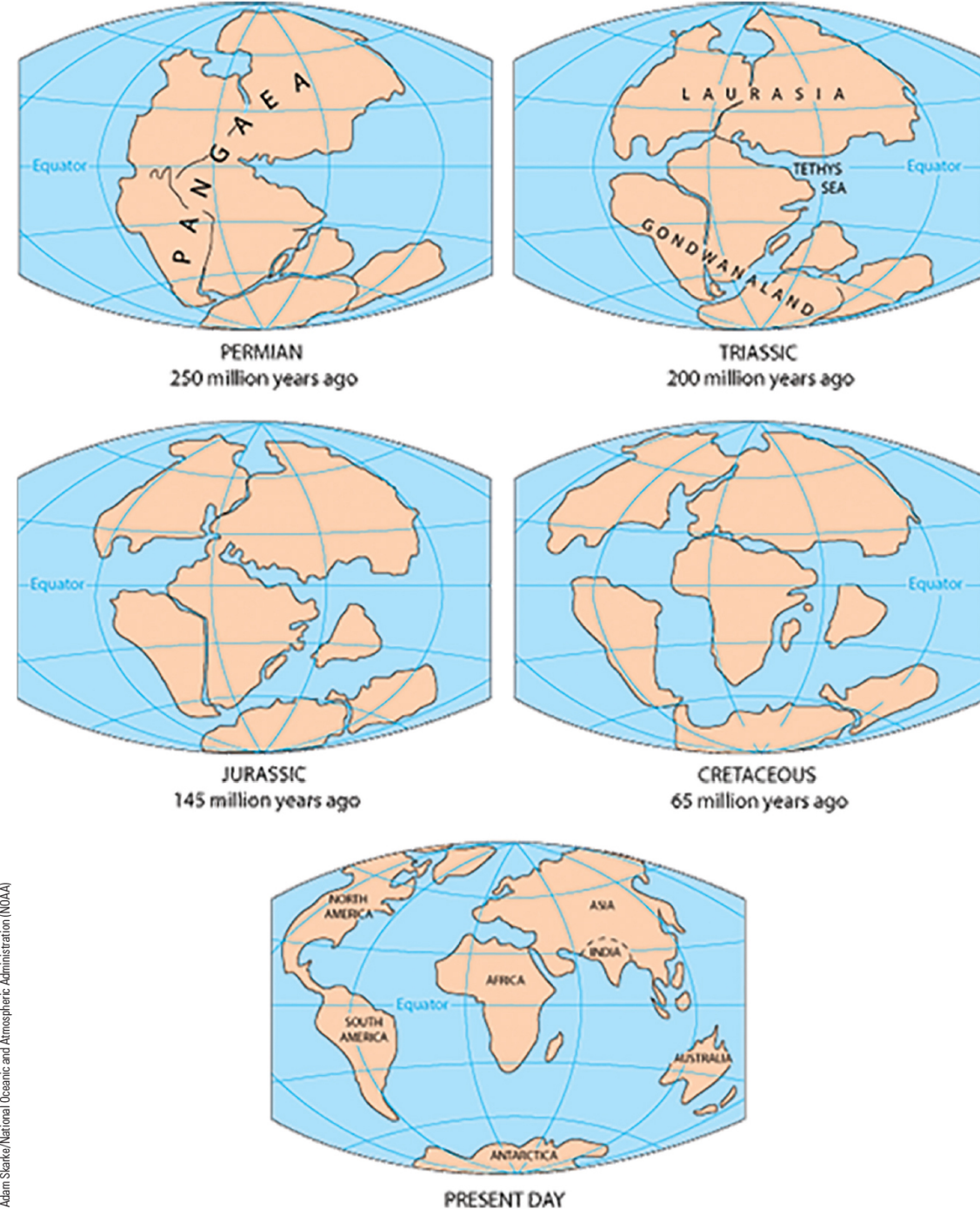


Figure 3.21 The breakup of Pangaea shown in five stages beginning about 250 million years ago.

Scientific Evidence Supporting Plate Tectonics Theory

Age of the Ocean Crust

Using radioactive dating and isotopic decay concept, scientist verified that younger crust is located along the mid-oceanic ridges.

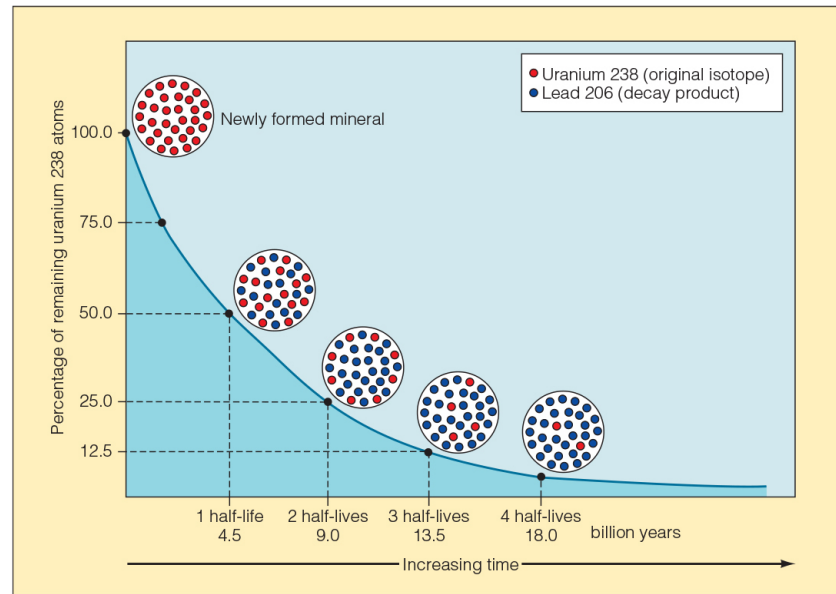


Figure A The rate of radioactive decay of a radioactive form of uranium (^{238}U) into lead. The half-life of ^{238}U is 4.5 billion years. During each half-life, one-half of the remaining amount of the radioactive element decays to become a different element. The assumption is made that the system has remained closed—no radioactive atoms or decay products have been added or removed from the sample. Ages obtained from absolute and relative dating of continental rocks and rocks and sediments from the seabeds coincide with ages predicted by plate tectonic theory.

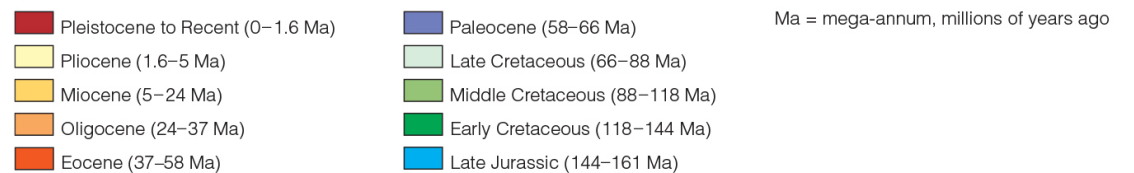
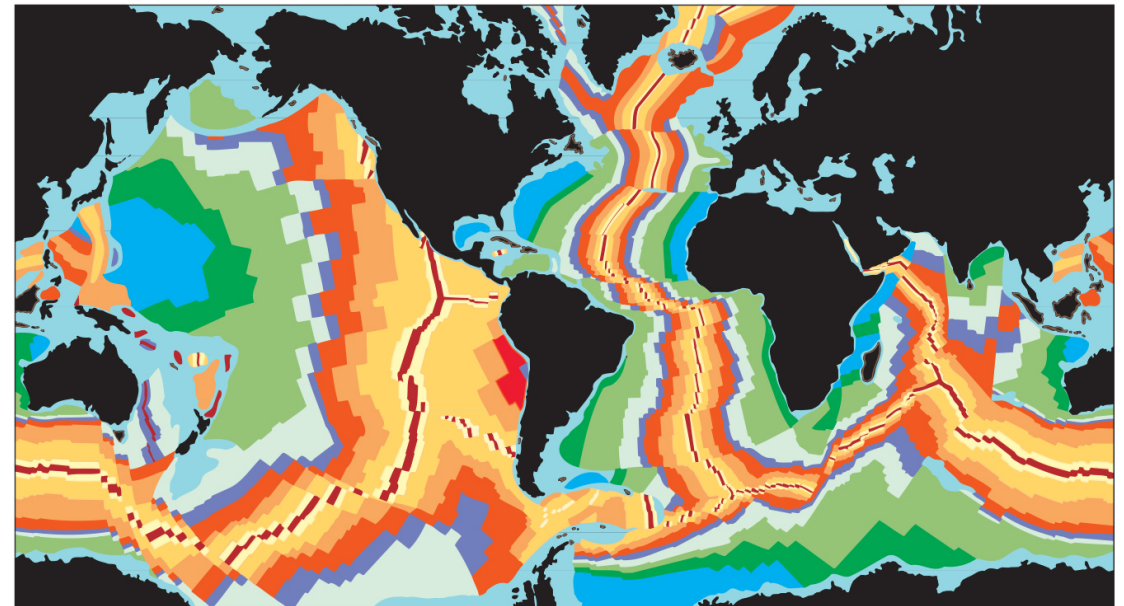


Figure 3.29 The age of the ocean floors. Colors represent an expression of seafloor spreading over the last 200 million years as revealed by paleomagnetic patterns. Note especially the relative symmetry of the Atlantic basin in contrast with the asymmetrical Pacific, where the spreading center is located close to the eastern margin and intersects the coast of California.

Thinking moment

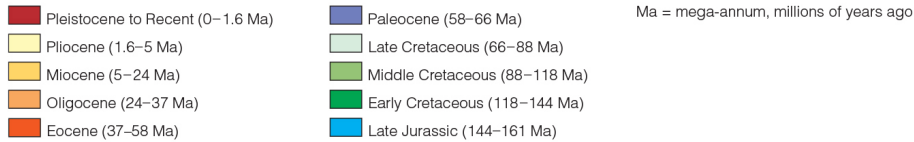
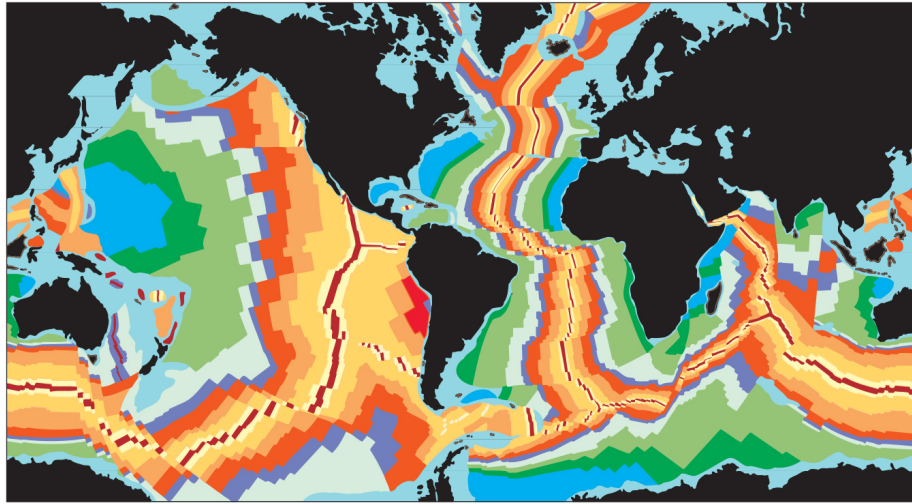


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Based on the fact we know the crust is moving over the Mantle due to convective Mantle motions.

Why do you think young crust are in the middle of the ocean basins along submarine mountain ridges?

- ☐ Because the continents are converging towards these mountains, closing the ocean basins
- ☐ Because the continents are diverging away from these mountains, opening the ocean basins
- ☐ Because crust is being destroyed by the mantle near these mountains
- ☐ Because the crust is being created along these mountain ridges

Seismic events are most common along the plates

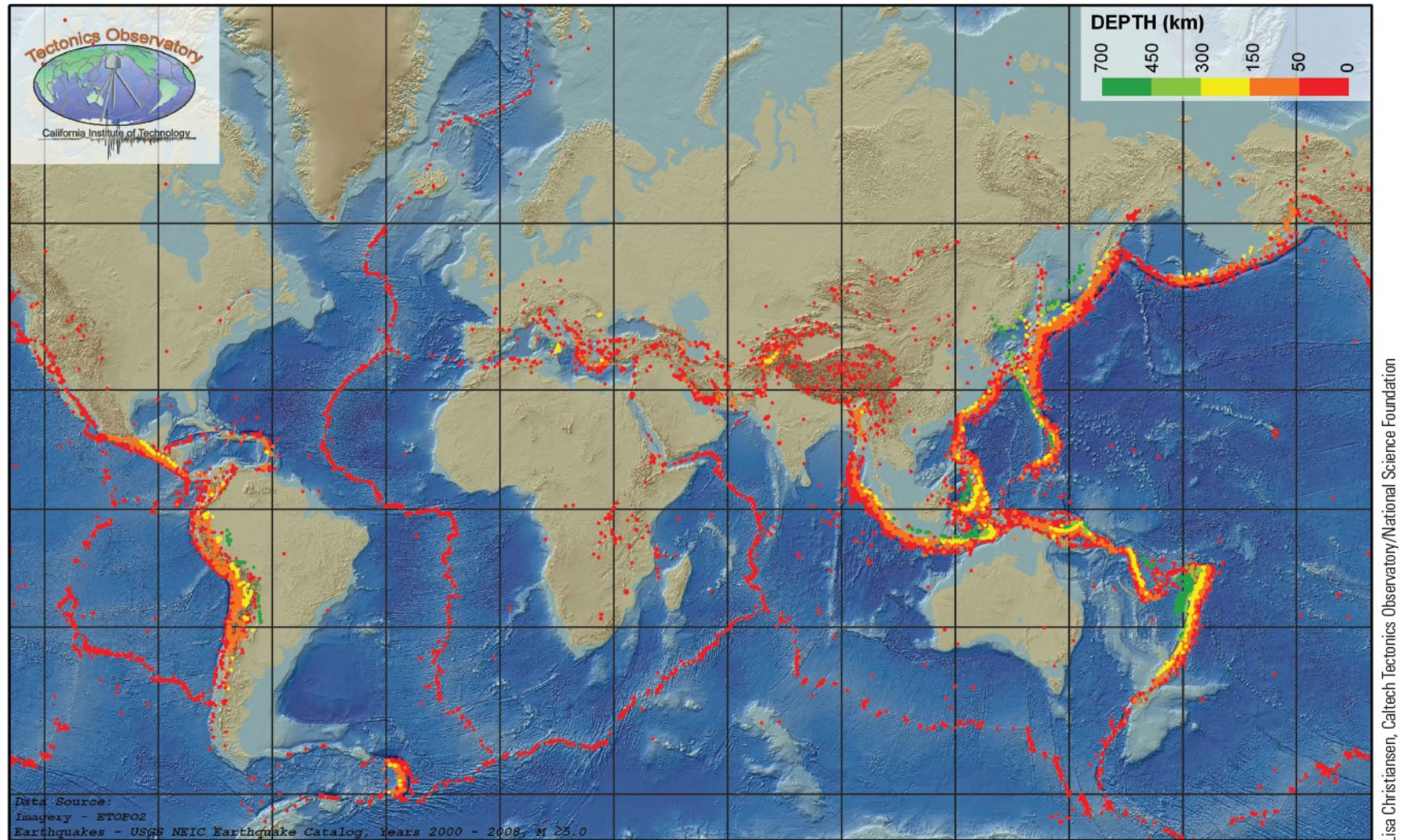


Figure 3.15 Seismic events, January, 2000, through December, 2008 (magnitude 5.0 M and above worldwide). Earthquake locations are depicted in red, orange, yellow, light green, and darker green to represent event depths of 0 to 50 kilometers (0–31 miles), 50 to 150 kilometers (31–93 miles), 150 to 300 kilometers (93–186 miles), 300 to 450 kilometers (186–280 miles), and 450 to 700 kilometers (280–435 miles), respectively.

Ocean floor mapping

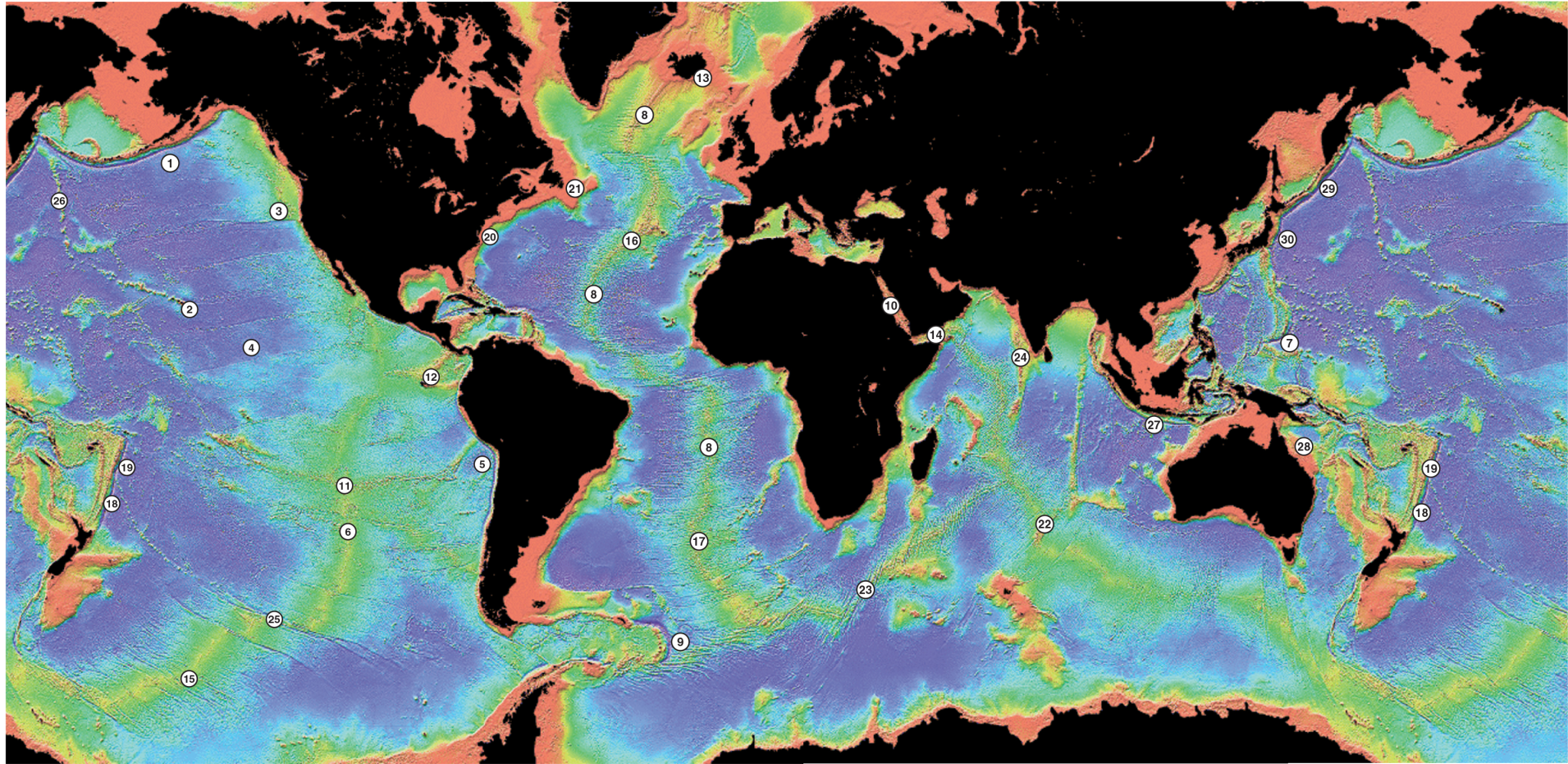


Figure 4.31 The Grand Tour. A technological *tour de force*: a map that shows all the features discussed in this chapter, derived from data provided to the National Geophysical Data Center from satellites and shipborne sensors. These features—and a basic understanding of the geological reasons for their existence—will help you recall the dramatic nature and history of the seafloor that we have discussed in the past two chapters.

Key to features:

- | | | | |
|-------------------------------------|------------------------------|-----------------------------|----------------------------|
| (1) Aleutian Trench | (9) South Sandwich Trench | (17) Tristan de Cunha | (25) Eltanin Fracture Zone |
| (2) Hawaiian Islands | (10) Red Sea | (18) Kermadec Trench | (26) Emperor Seamounts |
| (3) Juan de Fuca Ridge | (11) Easter Island | (19) Tonga Trench | (27) Java Trench |
| (4) Clipperton Fracture Zone | (12) Galápagos Rift | (20) Hatteras Abyssal Plain | (28) Great Barrier Reef |
| (5) Peru–Chile Trench | (13) Iceland | (21) Grand Bank | (29) Kuril Trench |
| (6) East Pacific Rise | (14) Gulf of Aden | (22) Mid-Indian Ridge | (30) Japan Trench |
| (7) Mariana Trench, Challenger Deep | (15) Pacific–Antarctic Ridge | (23) Atlantic–Indian Ridge | |
| (8) Mid-Atlantic Ridge | (16) Azores | (24) Maldives | |

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